

Uterine Corpus Endometrial Carcinoma (UCEC)

Simone Giudici - Bioinformatics A.Y. 2024-2025

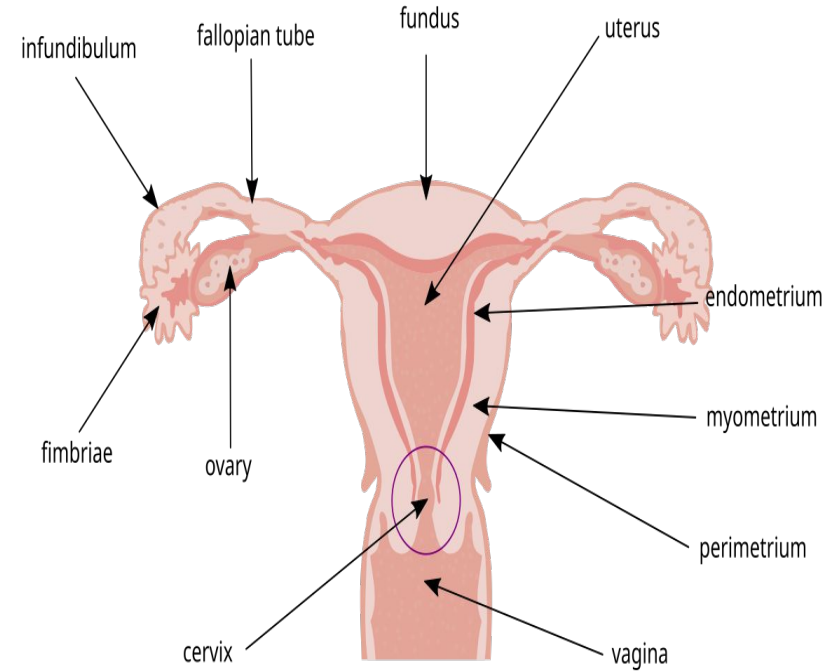
What is UCEC?

UCEC (Uterine Corpus Endometrial Carcinoma) is the most common form of **endometrial cancer**, arising from the **epithelial cells** lining the uterus (endometrium). Because it originates in epithelial tissue, it is classified as a **carcinoma**—a type of cancer that starts in the cells covering internal and external body surfaces.

UCEC is usually detected early due to symptoms like **abnormal vaginal bleeding**, especially in postmenopausal women.

There are two major types:

- **Type I (Endometrioid):**
Estrogen-dependent, typically low-grade, and slower-growing. It accounts for the majority of cases and is associated with a better prognosis. Often linked to hormonal imbalances, obesity, or unopposed estrogen exposure.
- **Type II (Non-Endometrioid):**
Estrogen-independent, often high-grade, and more aggressive. This includes **serous and clear cell carcinomas**, which are less common but carry a higher risk of spread and recurrence. These are not typically linked to hormonal factors and have a poorer prognosis.



The dataset

The analysis I performed is based on a TCGA (The Cancer Genome Atlas Program) RNA sequencing dataset.

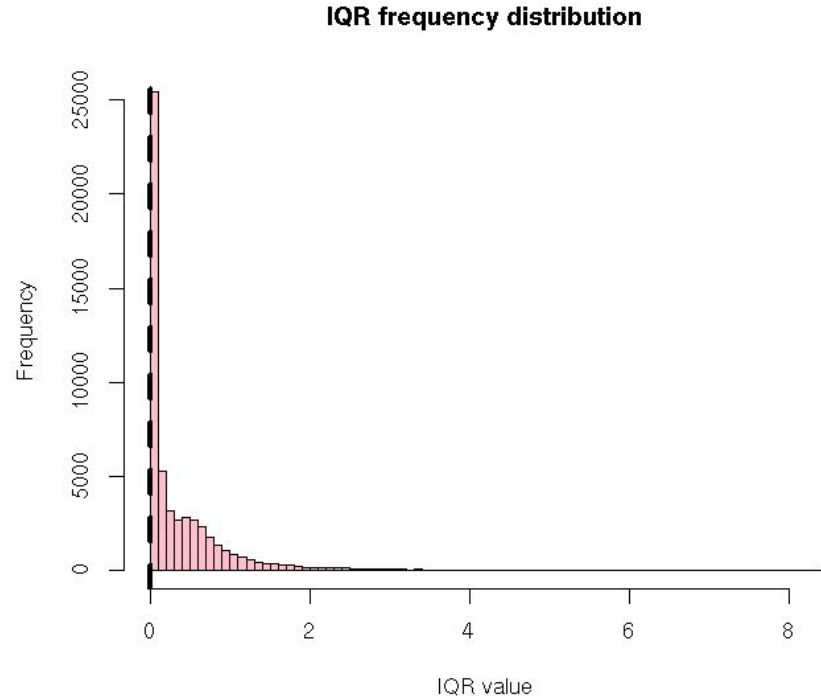
The dataset consists of a two-dimensional matrix: the **rows represent gene symbols**, the **columns represent samples** (in the form of **TCGA barcodes**), and the values correspond to the expression level of a gene in the respective sample.

Along with the said matrix , a “ucec_clinical.txt” file containing clinical evaluation data for each patient was provided. This data turned out to be crucial for the final analysis so it is going to be better mentioned later in the presentation.

The analysis / Pre-processing

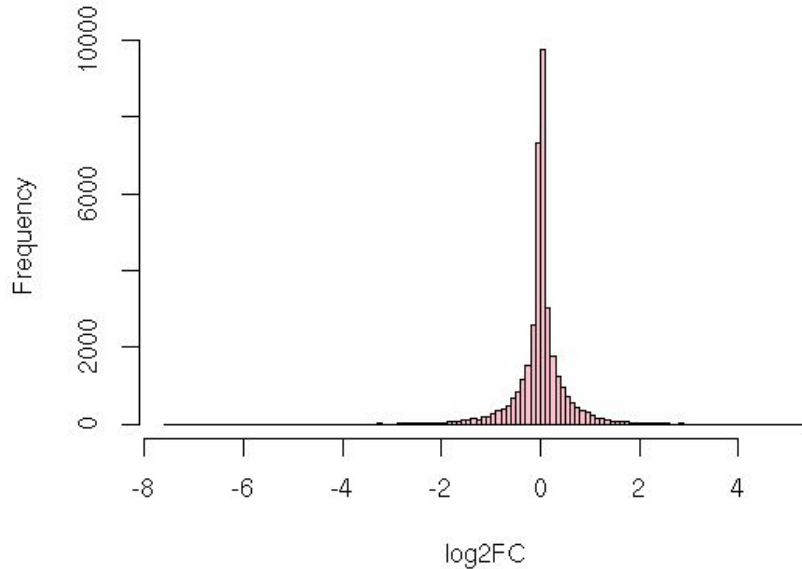
The data, as extracted from the matrix, is not suitable for analysis and needs a pre-processing phase. In particular, the following steps were taken:

1. Removed patients with no “paired” samples (a sample from cancerous tissue and one from adjacent healthy tissue)
2. Removed genes whose overall expression mean turned out to be 0
3. Applied a logarithmic normalization
4. Removed genes with IQR < 10th percentile



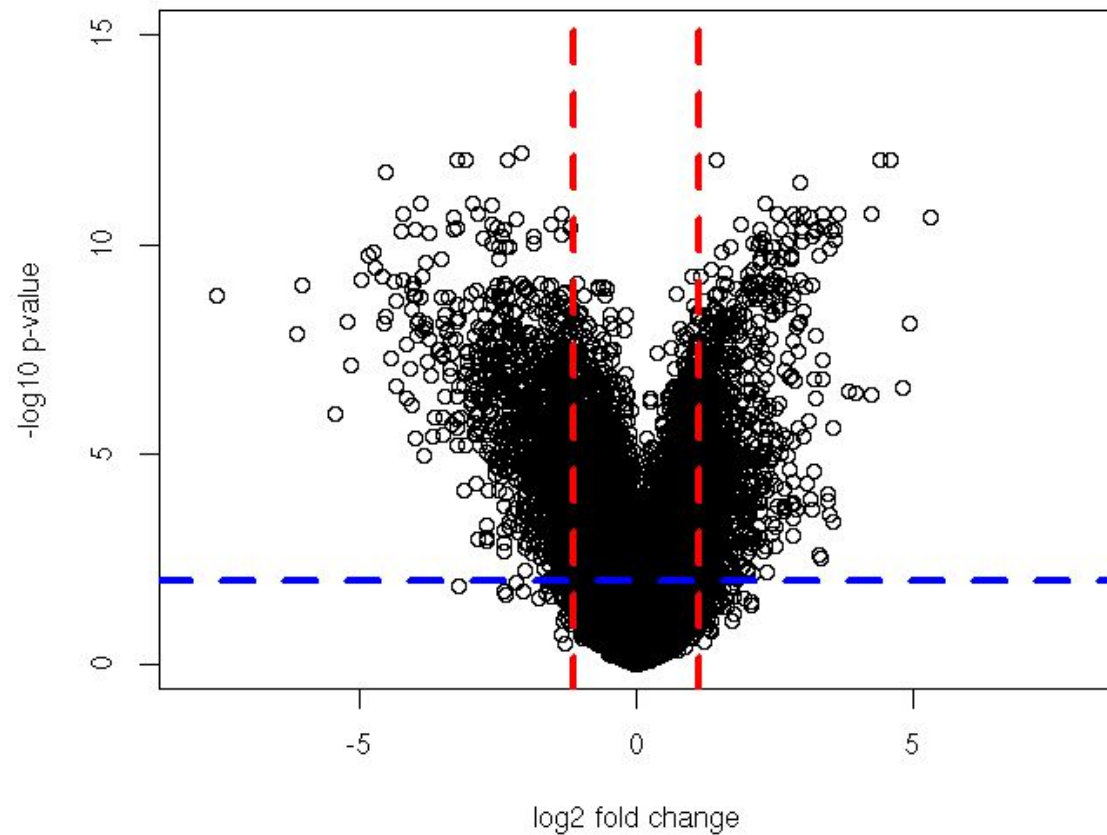
The analysis / Filtering

FC (logarithmic) frequency distribution



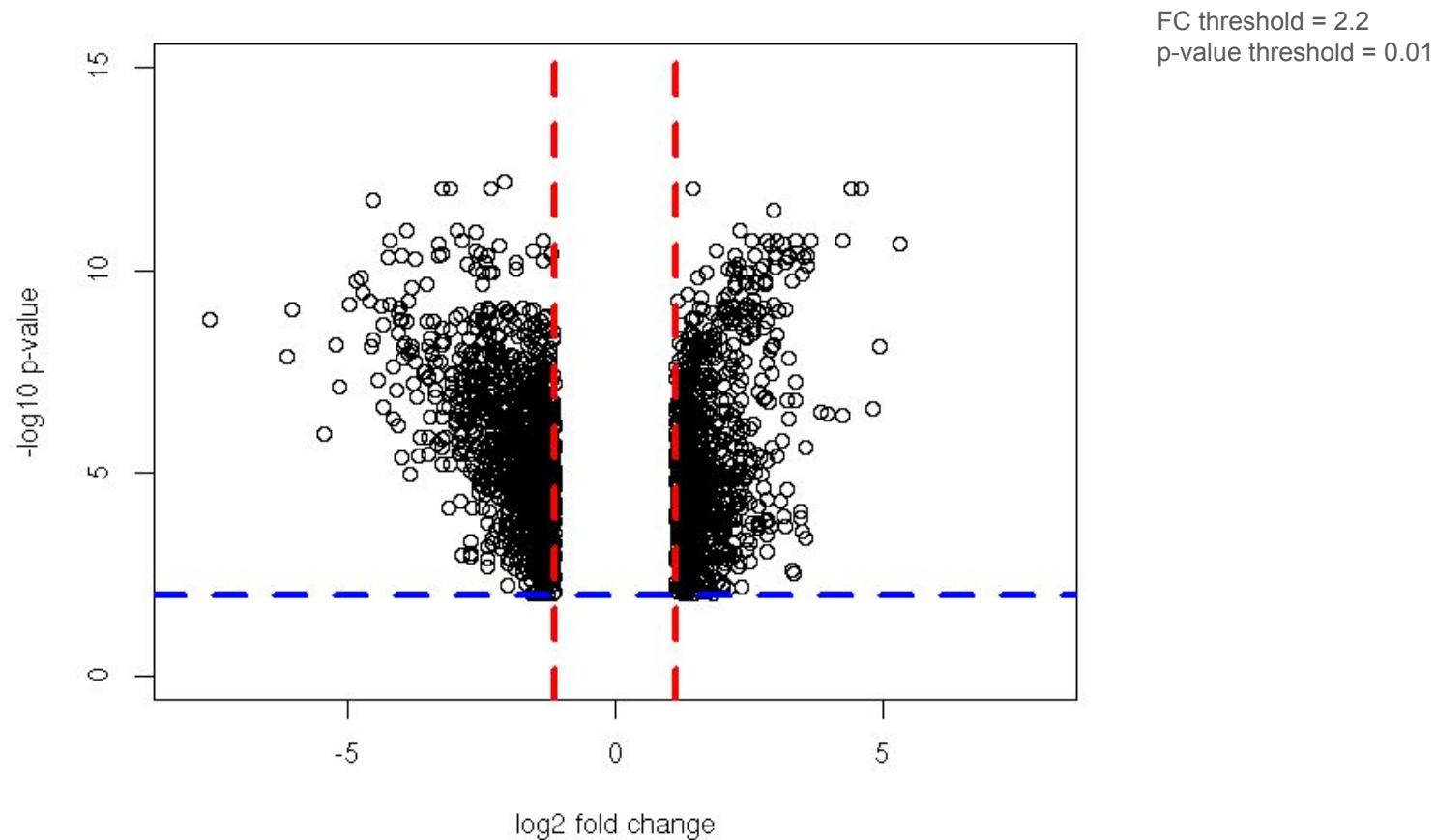
At this point the logarithmic fold change and the p-value were calculated. Since the sample size $n > 30$ I exploited the hypothesis that the distribution was normal, hence I applied the paired t-test. The p-values were adjusted using the false discovery rate technique. A volcano plot was then created according to the newly obtained values.

volcano plot before filtering



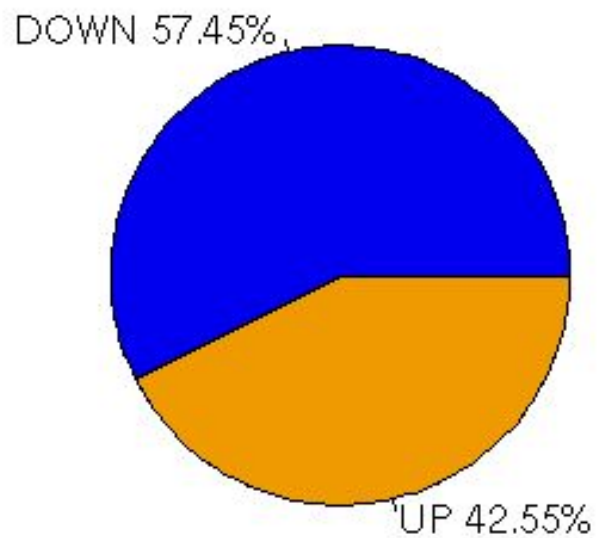
FC threshold = 2.2
p-value threshold = 0.01

volcano plot after filtering

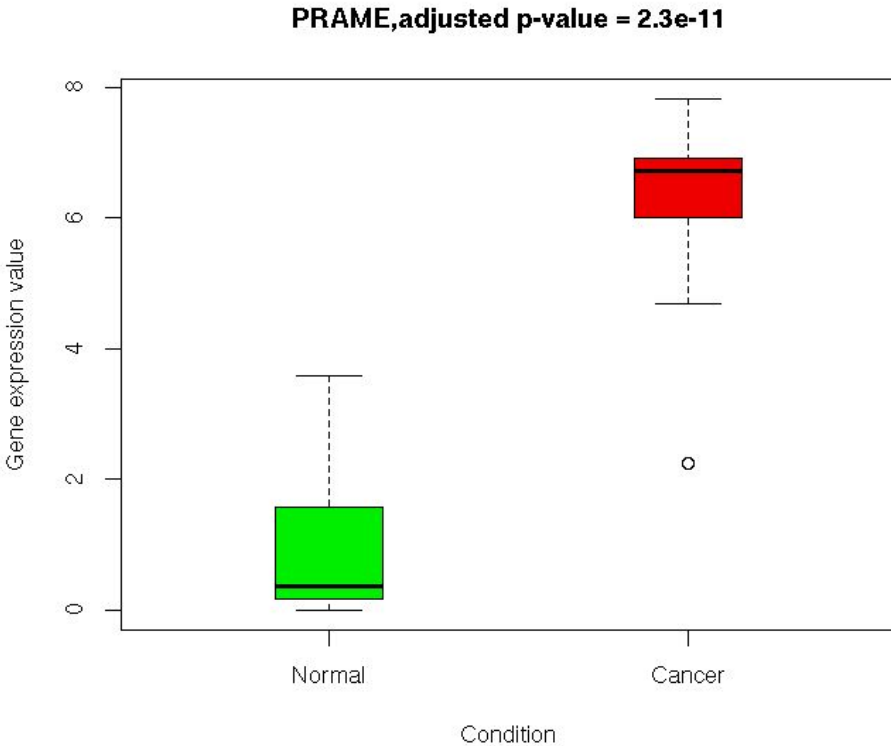


60661 -> 2061

589 -> 46



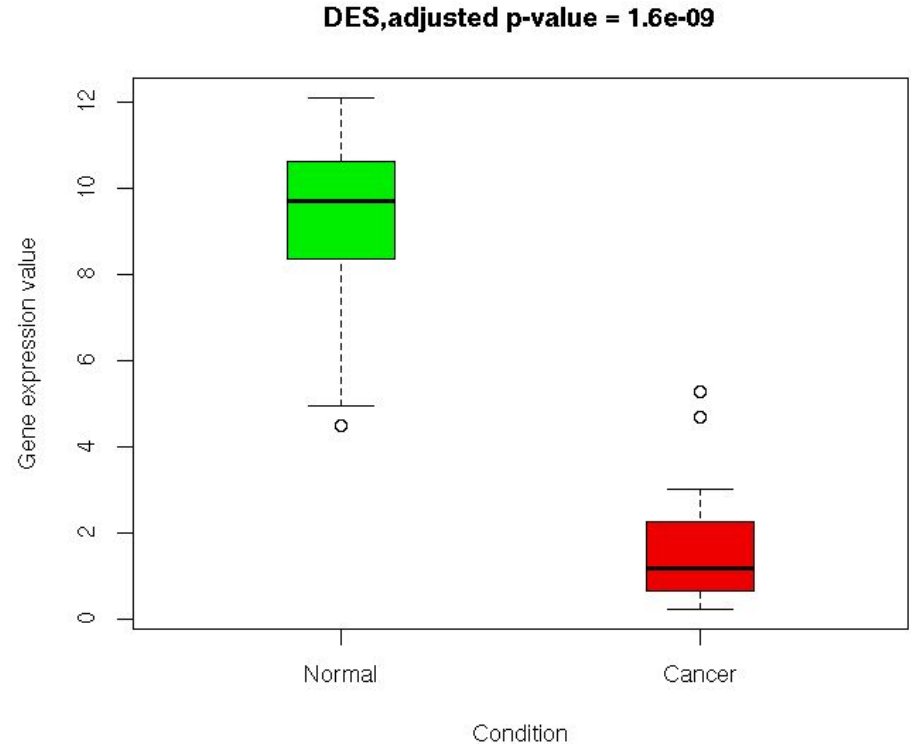
Most up-regulated gene

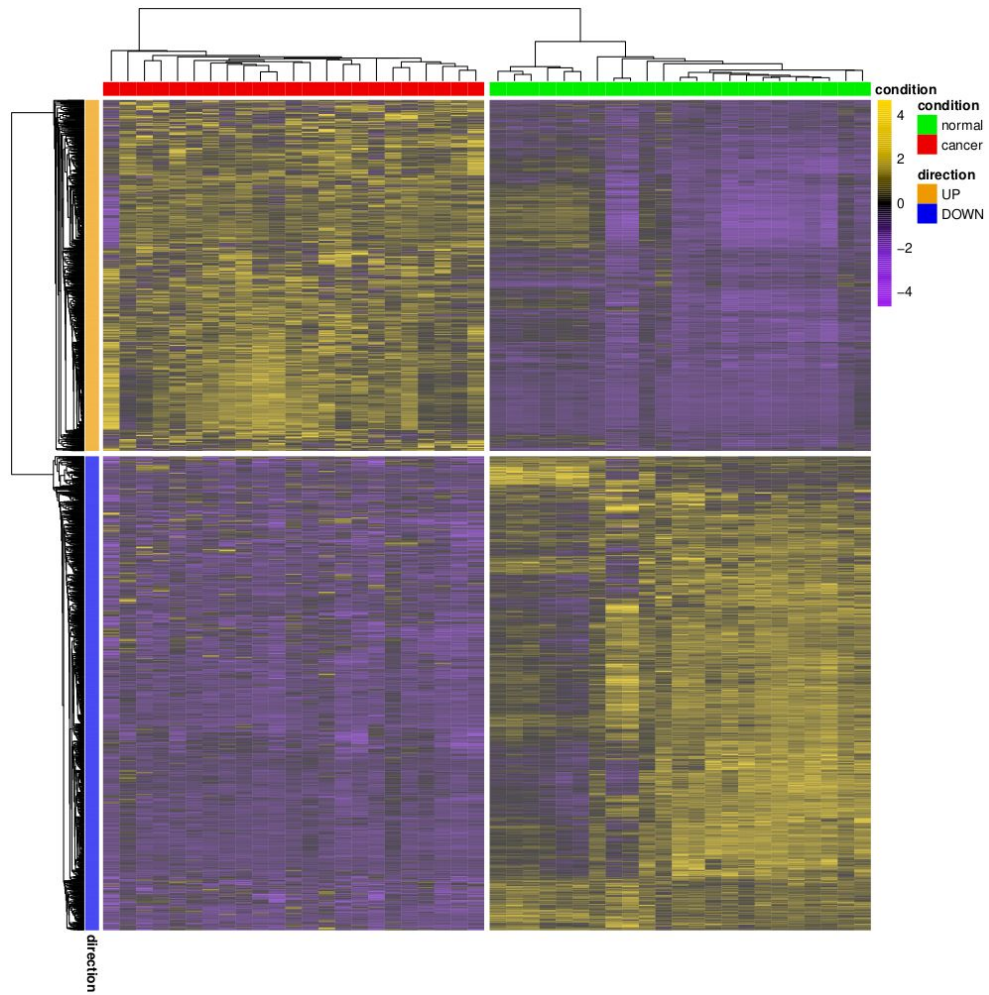


This gene encodes an antigen that is preferentially expressed in human melanomas and that is recognized by cytolytic T lymphocytes. It is not expressed in normal tissues, except testis. The encoded protein acts as a repressor of retinoic acid receptor, and likely confers a growth advantage to cancer cells via this function.

Most down-regulated gene

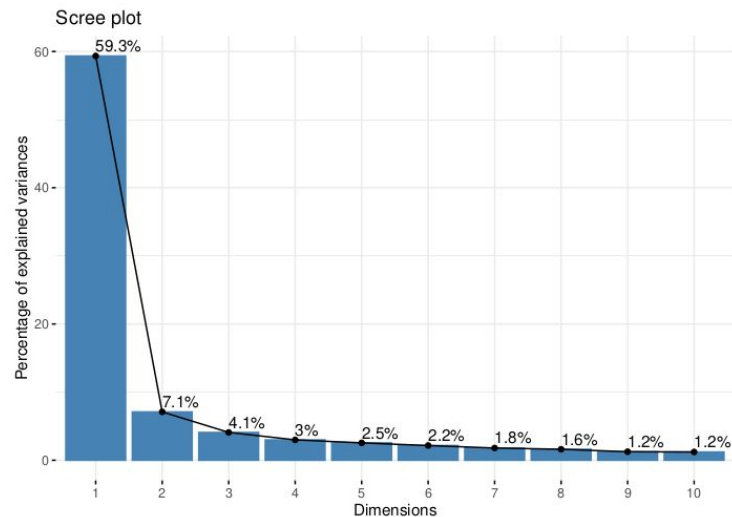
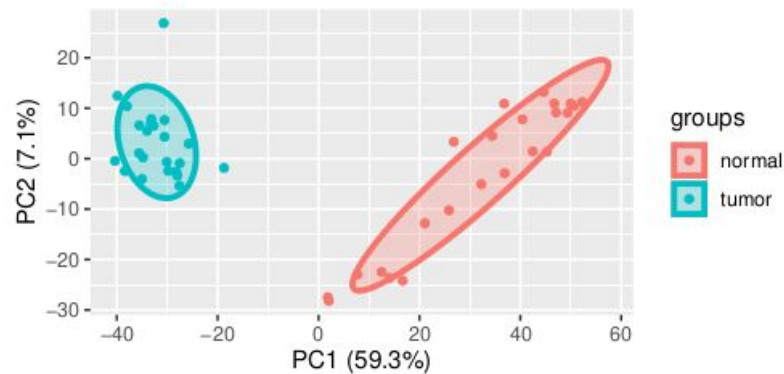
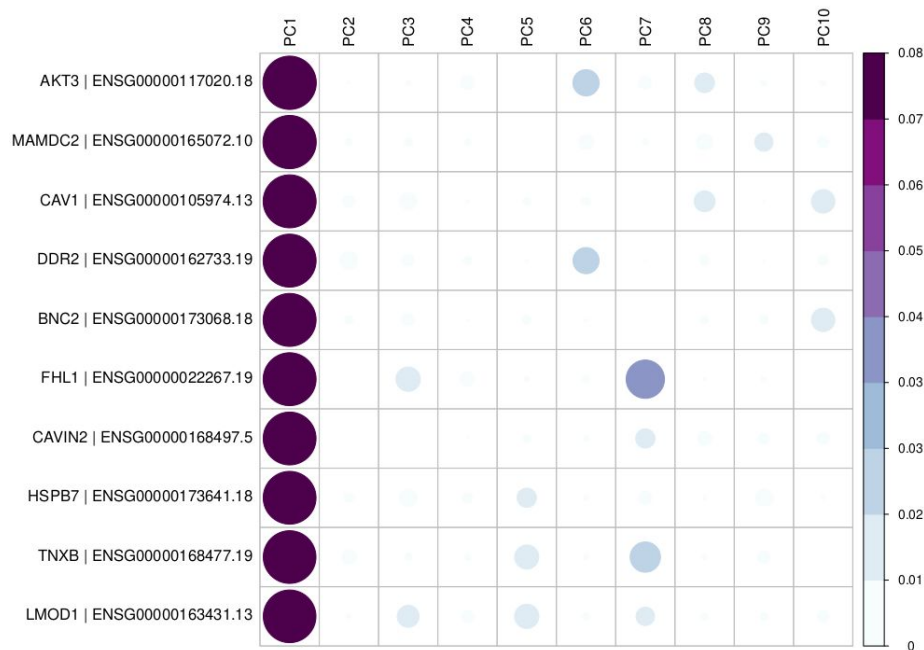
This gene encodes a muscle-specific class III intermediate filament. Homopolymers of this protein form a stable intracytoplasmic filamentous network connecting myofibrils to each other and to the plasma membrane. Mutations in this gene are associated with desmin-related myopathy, a familial cardiac and skeletal myopathy (CSM), and with distal myopathies.



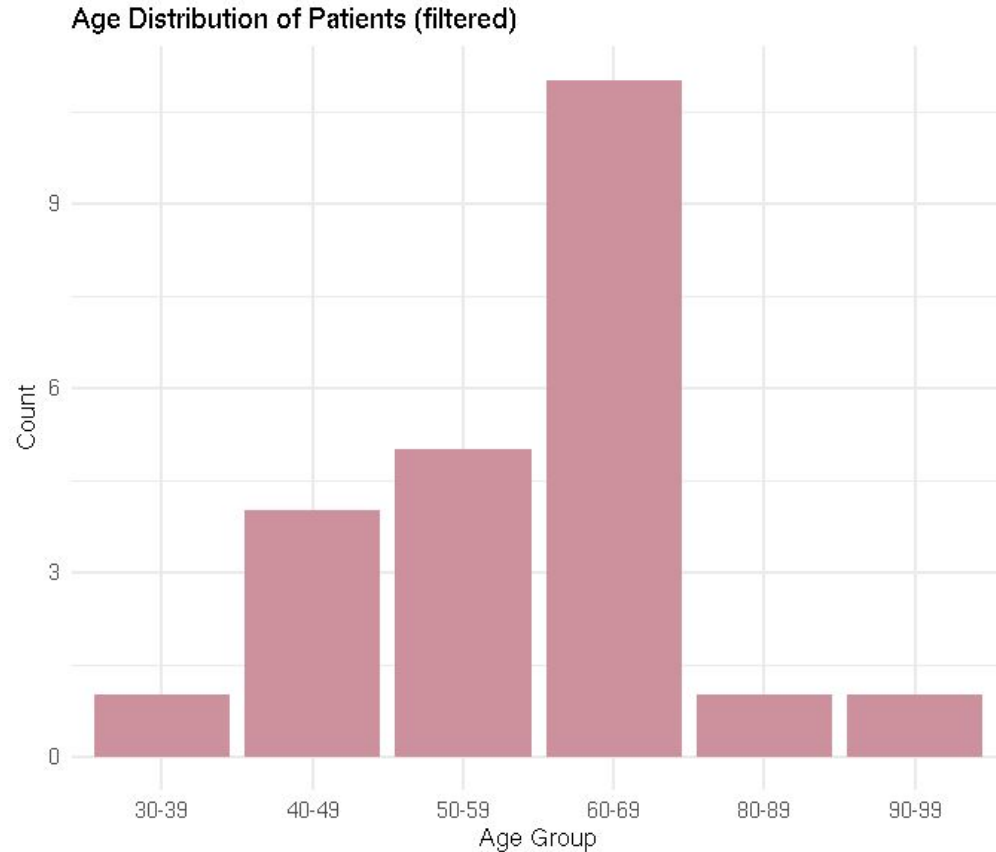


Heatmap

PCA analysis

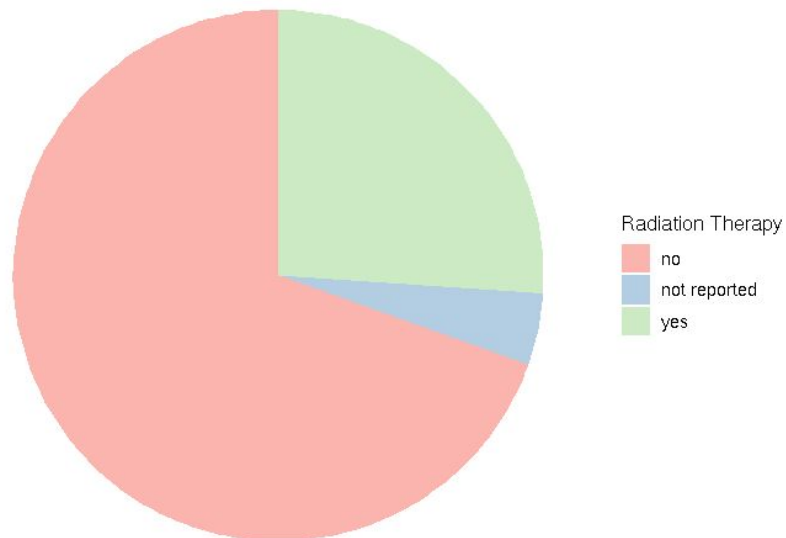


Clinical data after filtering/age distribution

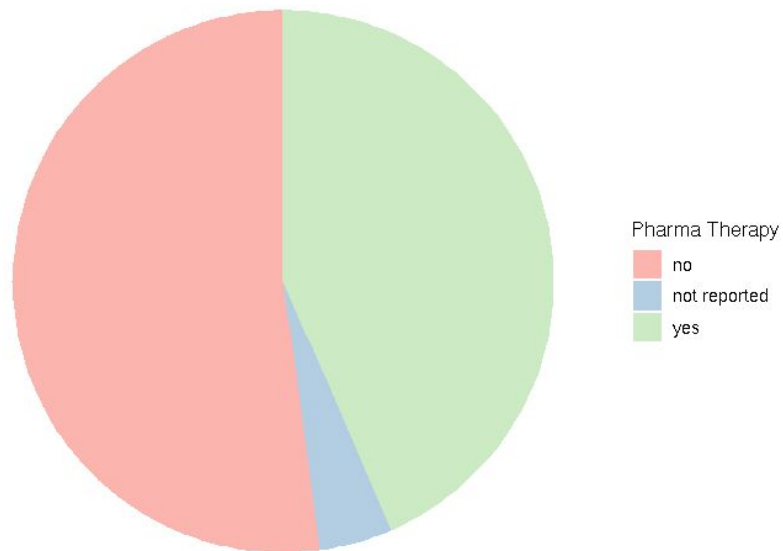


Clinical data after filtering/treatment

Radiation Therapy Distribution



Pharmaceutical Therapy Distribution



Biomarkers

*A **biomarker** is a biological molecule found in blood, other body fluids, or tissues that is a sign of a normal or abnormal process, or of a condition or disease. A biomarker may be used to see how well the body responds to a treatment for a disease or condition. Also called molecular marker and signature molecule.**

As the analysis was performed, it was possible to spot some biomarkers among the roughly 2000 genes resulting from the filtering, whose role was later confirmed by a literature-based research.

* <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/biomarker>

Biomarkers/MCM10

Gene Symbol	MCM10
Gene ID	55388
Expression direction	Upregulated
Location	10p13
Literature search	Chen J, Wu S, Wang J, Han C, Zhao L, He K, Jia Y, Cui M. MCM10: An effective treatment target and a prognostic biomarker in patients with uterine corpus endometrial carcinoma. J Cell Mol Med. 2023 Jun;27(12):1708-1724. doi: 10.1111/jcmm.17772. Epub 2023 May 29. PMID: 37246638; PMCID: PMC10273062.

Biomarkers/SHCBP1

Gene Symbol	SHCBP1
Gene ID	79801
Expression direction	Upregulated
Location	16q11.2
Literature search	Wang N, Zhu L, Wang L, Shen Z, Huang X. Identification of SHCBP1 as a potential biomarker involving diagnosis, prognosis, and tumor immune microenvironment across multiple cancers. Comput Struct Biotechnol J. 2022 Jun 18;20:3106-3119. doi: 10.1016/j.csbj.2022.06.039. PMID: 35782736; PMCID: PMC9233189.

Biomarkers/UBE2T

Gene Symbol	UBE2T
Gene ID	29089
Expression direction	Upregulated
Location	1q32.1
Literature search	Wang L, Wang M, Wang Z, Wang K, Zhao B, Wang Y, Zheng J, Zhang S. UBE2T is a diagnostic and prognostic biomarker for endometrial cancer. Clin Transl Oncol. 2025 May;27(5):2067-2083. doi: 10.1007/s12094-024-03713-z. Epub 2024 Oct 5. PMID: 39367897; PMCID: PMC12033108.

Biomarkers/APOA1

Gene Symbol	APOA1
Gene ID	335
Expression direction	Downregulated
Location	11q23.3
Literature search	Xu P, Zhang Q, Zhai J, Chen P, Deng X, Miao L, Zhang X. APOA1 promotes tumor proliferation and migration and may be a potential pan-cancer biomarker and immunotherapy target. Transl Oncol. 2025 May;55:102344. doi: 10.1016/j.tranon.2025.102344. Epub 2025 Mar 14. PMID: 40088749; PMCID: PMC11957500.

Biomarkers/CCNE1

Gene Symbol	CCNE1
Gene ID	898
Expression direction	Upregulated
Location	19q12
Literature search	Zheng X, Chen L, Liu W, Zhao S, Yan Y, Zhao J, Tian W, Wang Y. CCNE1 is a predictive and immunotherapeutic indicator in various cancers including UCEC: a pan-cancer analysis. Hereditas. 2023 Mar 24;160(1):13. doi: 10.1186/s41065-023-00273-0. PMID: 36964635; PMCID: PMC10037856.

Biomarkers/Conclusions

The literature-based research first of all confirmed the results obtained from my own expression analysis and secondly indicated that these biomarkers showed to be effective targets for therapy (by silencing a gene if upregulated and enhancing it if downregulated) and indicators for prognosis that is, in general, a forecast of the course of the disease.



microRNA

MicroRNA are small, single-stranded, non coding RNA molecules containing 21–23 nucleotides. miRNAs are involved in RNA silencing and post-transcriptional regulation of gene expression by base-pairing to complementary sequences in messenger RNA (mRNA) molecules.

miRNAs are abundant in many mammalian cell types. They appear to target about 60% of the genes of humans and other mammals. Many miRNAs are evolutionarily conserved, which implies that they have important biological functions.

microRNA

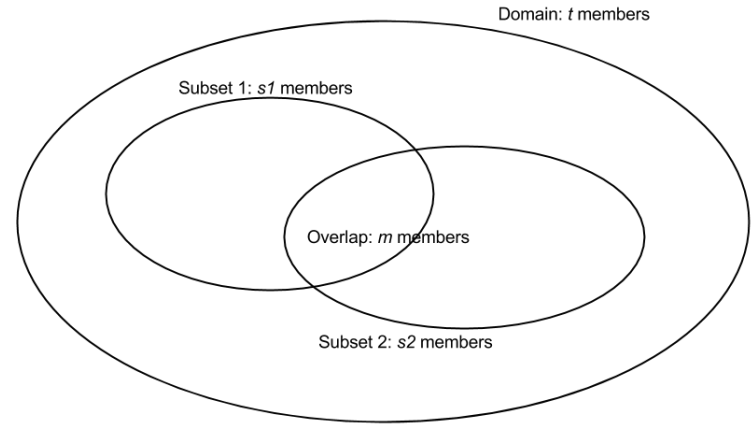
The following article *Lu X, Jing L, Liu S, Wang H, Chen B. miR-149-3p Is a Potential Prognosis Biomarker and Correlated with Immune Infiltrates in Uterine Corpus Endometrial Carcinoma. Int J Endocrinol. 2022 Jun 8;2022:5006123. doi: 10.1155/2022/5006123. PMID: 35719192; PMCID: PMC9200575.* (<https://pmc.ncbi.nlm.nih.gov/articles/PMC9200575/>) published on the **International Journal of Endocrinology** confirmed that the high expression of miRNA **miR-149-3p** (hence low expression of its target genes) is associated with a poor prognosis in UCEC.

None of the cited target genes was found to be upregulated in my analysis while three of them (**H1C1,INMT,SPRYD3**) were found among the downregulated genes thus confirming the interaction with the above miRNA.

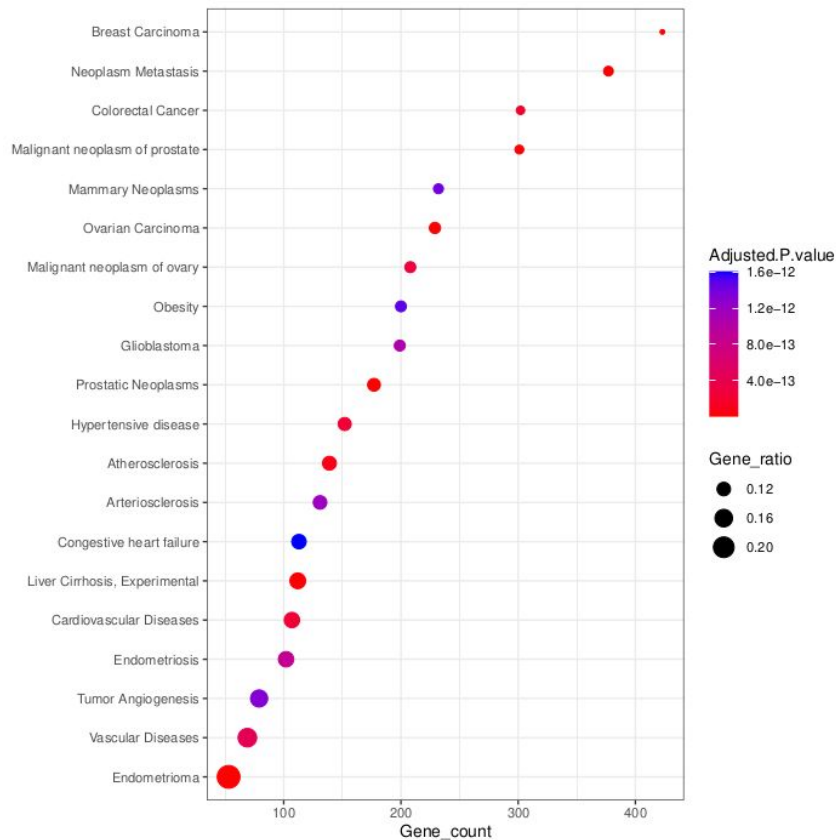
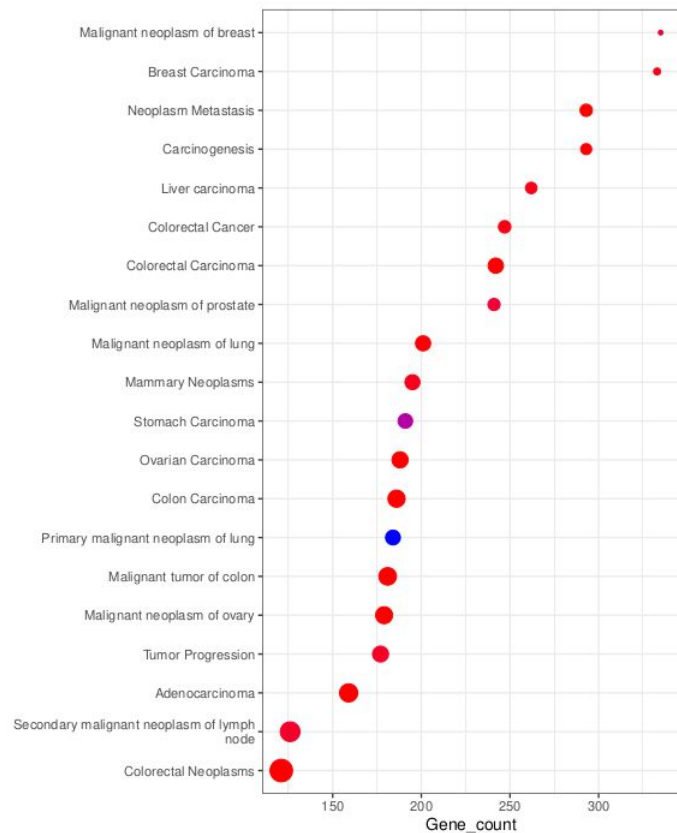
Functional enrichment

Functional enrichment is a method used in bioinformatics to determine whether a set of genes (such as differentially expressed genes) is significantly associated with known biological functions, pathways, or processes more than would be expected by chance.

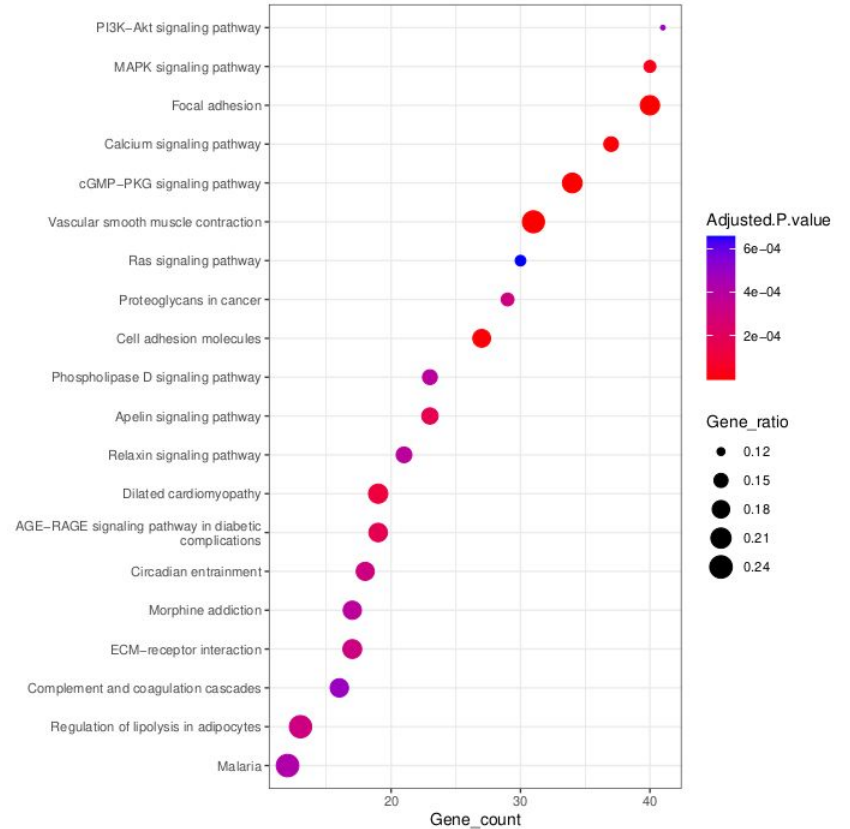
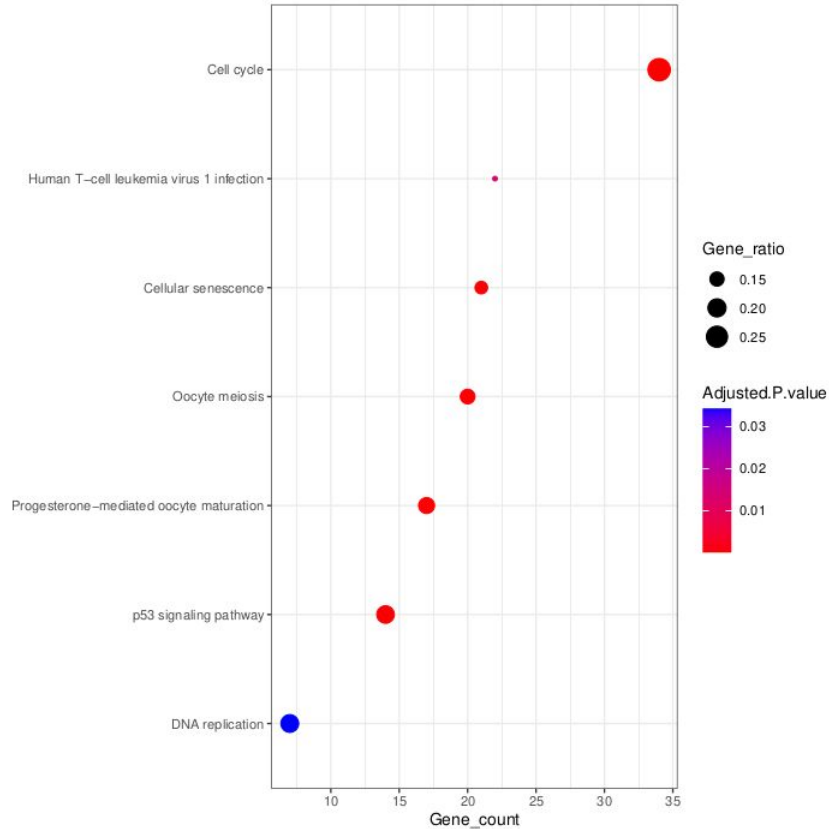
It's a statistical analysis based on the **Hypergeometric test** (also known as test of over-representation)



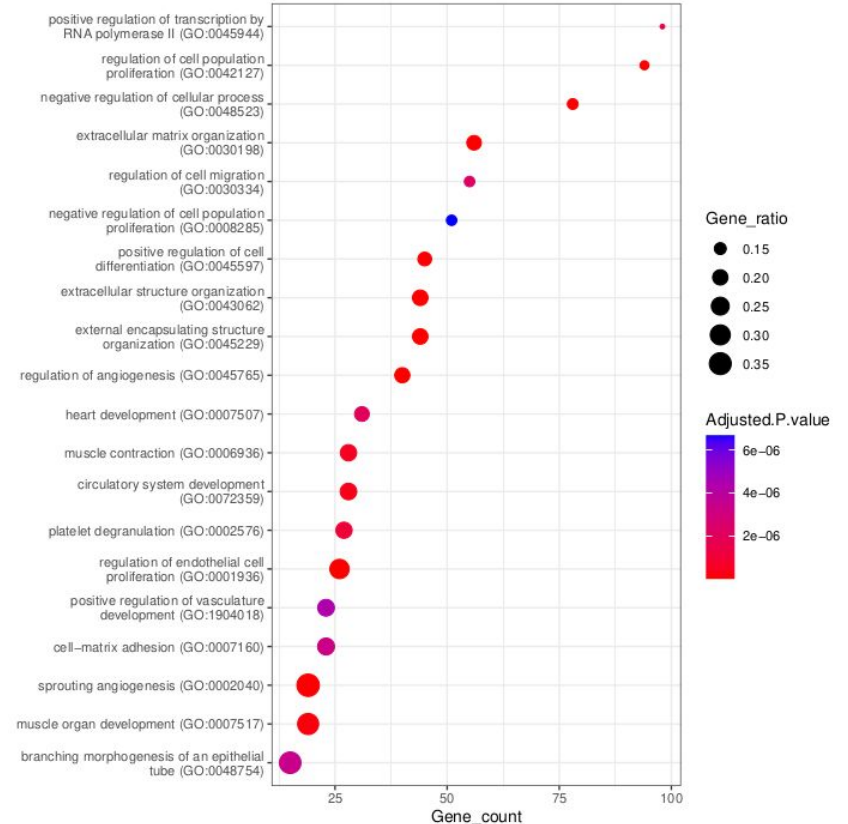
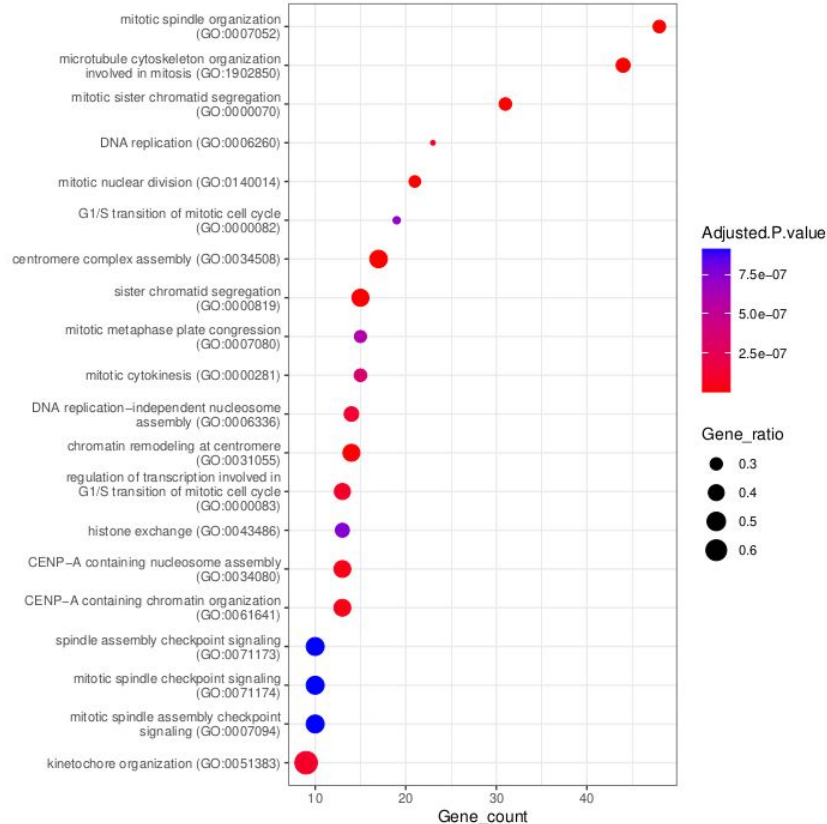
The enrichment/results/DisGeNET



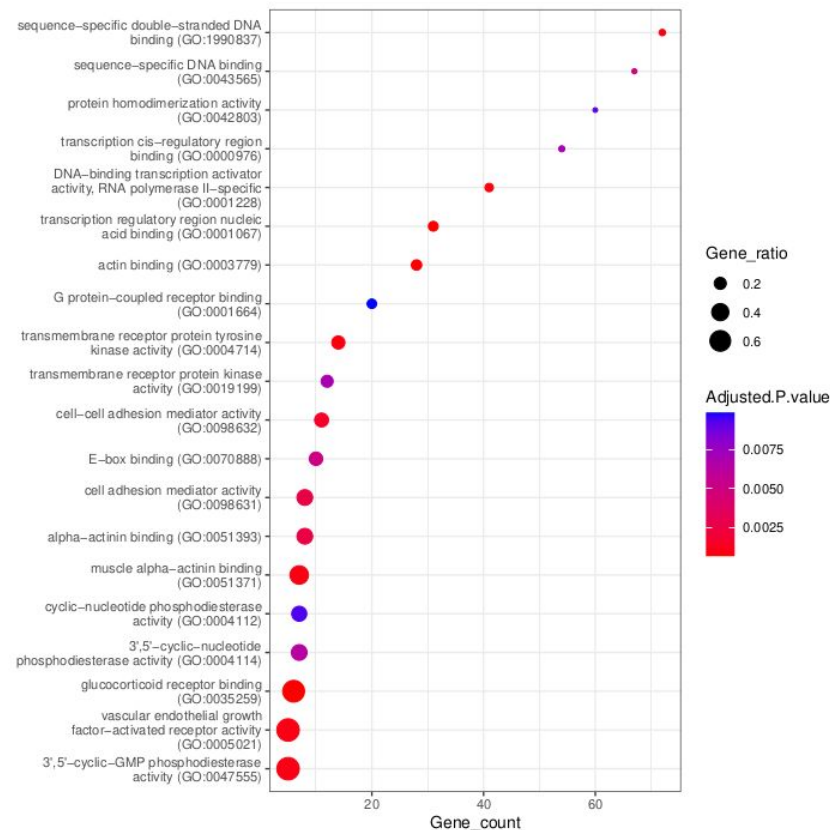
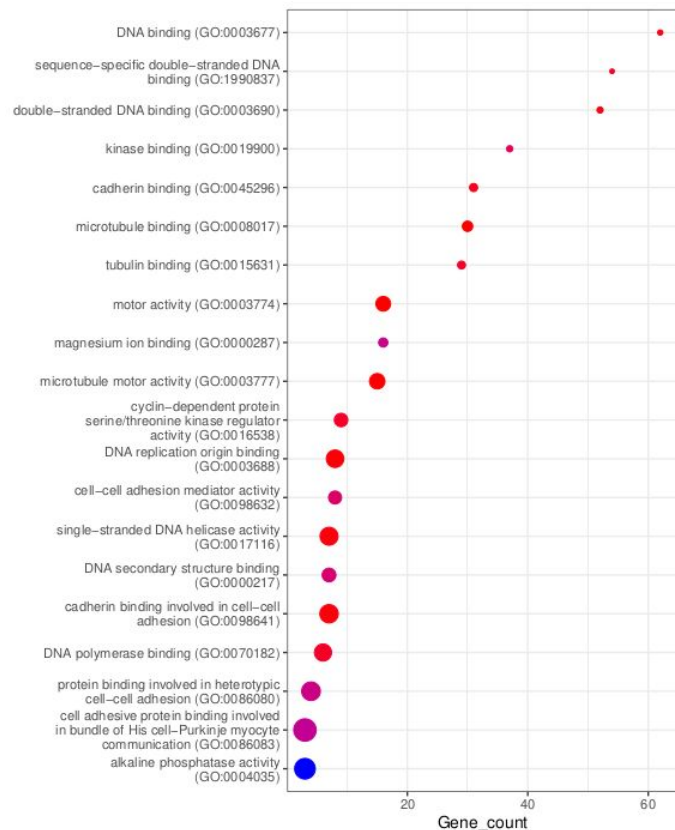
The enrichment/results/KEGG



The enrichment/results/GO biological process



The enrichment/results/GO molecular function



KEGG

KEGG (**Kyoto Encyclopedia of Genes and Genomes**) is a database used for understanding high-level functions of the biological system, such as the cell, the organism and the ecosystem, from molecular-level information, especially large-scale molecular datasets generated by genome sequencing.

A KEGG **pathway** is a manually drawn map that shows the interactions between genes in the context of metabolism, cellular processes and even human diseases.

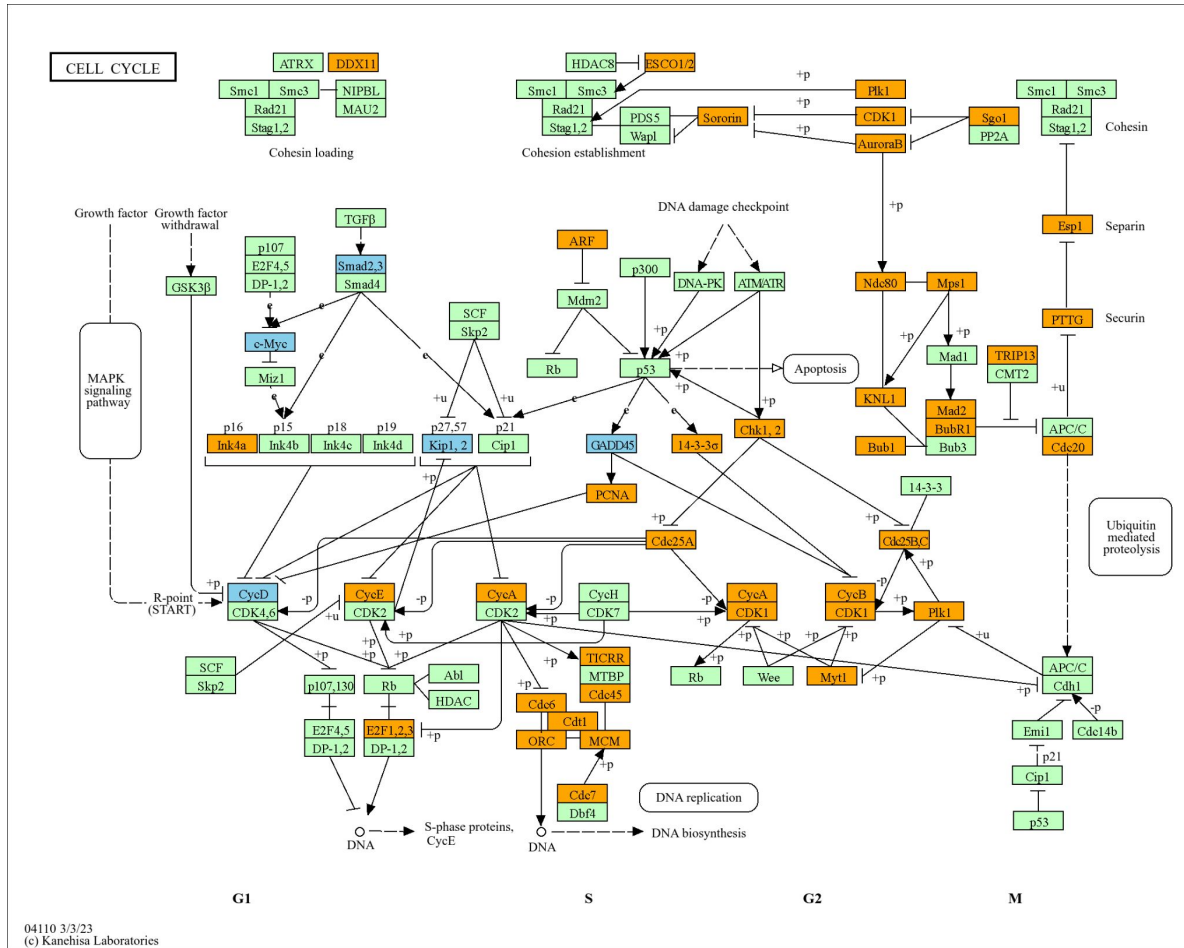
In the following slides some significant pathways coming both from the performed enrichment and from the previous-mentioned literature are reported.

KEGG

An important functionality of KEGG PATHWAY is the possibility to highlight some chosen genes using the format **<entrezID> \tab <colour>**. To do so I created a simple R script that converts gene symbols to entrezID and adds them in a list followed by the color **orange** if upregulated and **skyblue** if downregulated.

After exporting the list in a .txt file it was sufficient to add it to the KEGG portal to apply the filter.

KEGG/Cell cycle

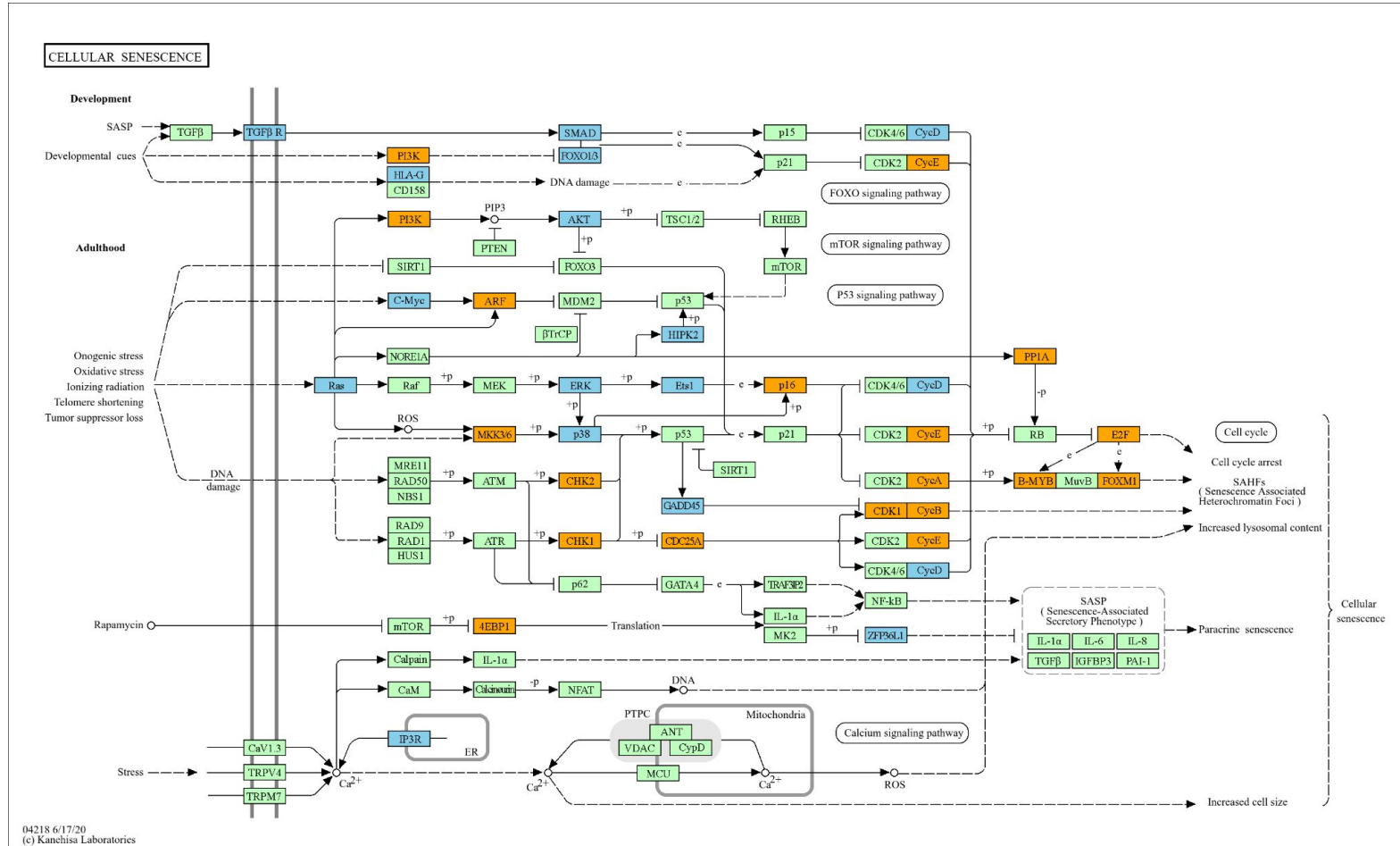


KEGG/Cell cycle

We can clearly see that a large part of genes involved in this pathway is upregulated. This typically indicates increased cellular proliferation, which is a hallmark of cancer and other hyperproliferative disorders.

In particular in the previous image we can see a “cluster” of genes involved in DNA replication as promoters that are upregulated, the ones that resulted to be downregulated act as inhibitors.

KEGG/Cellular senescence

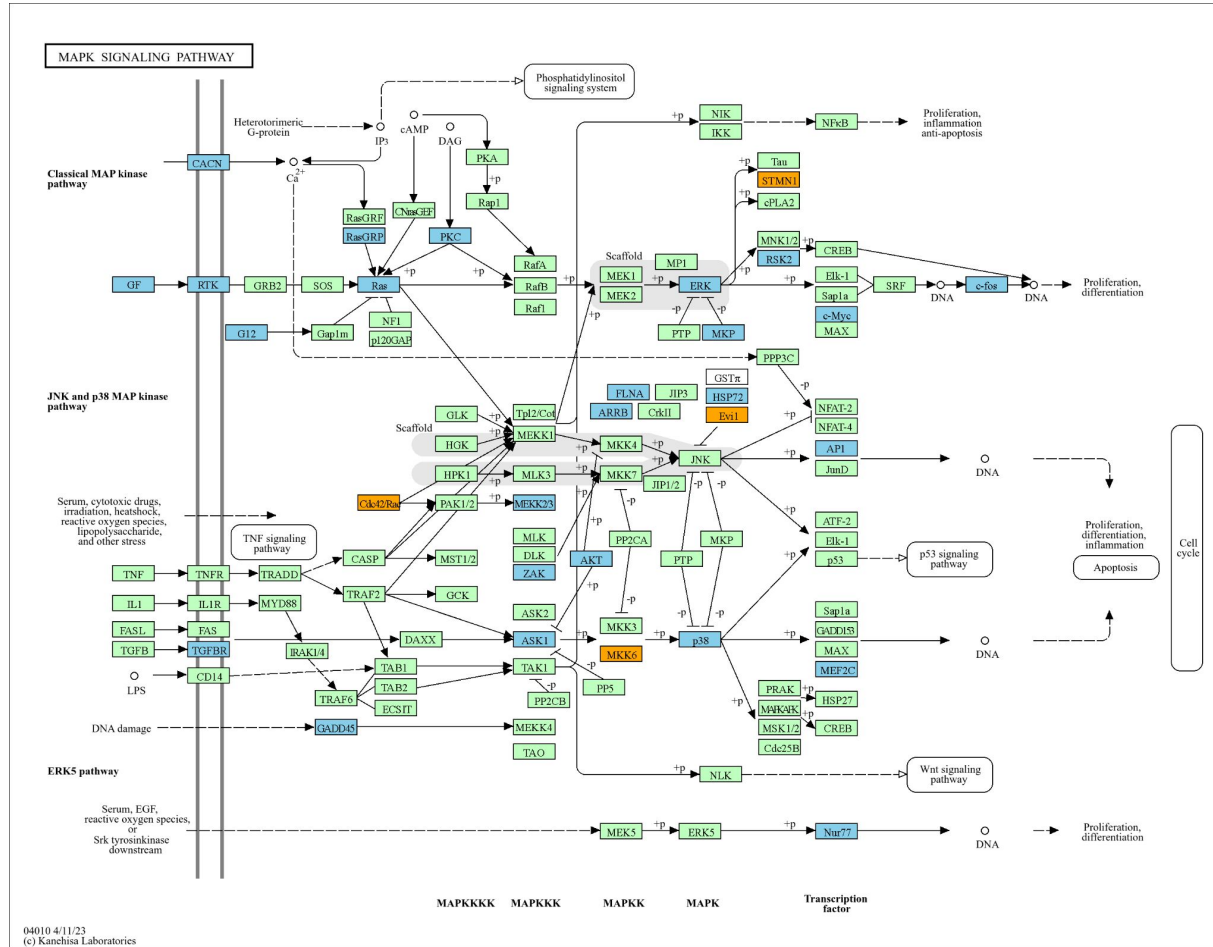


KEGG/Cellular senescence

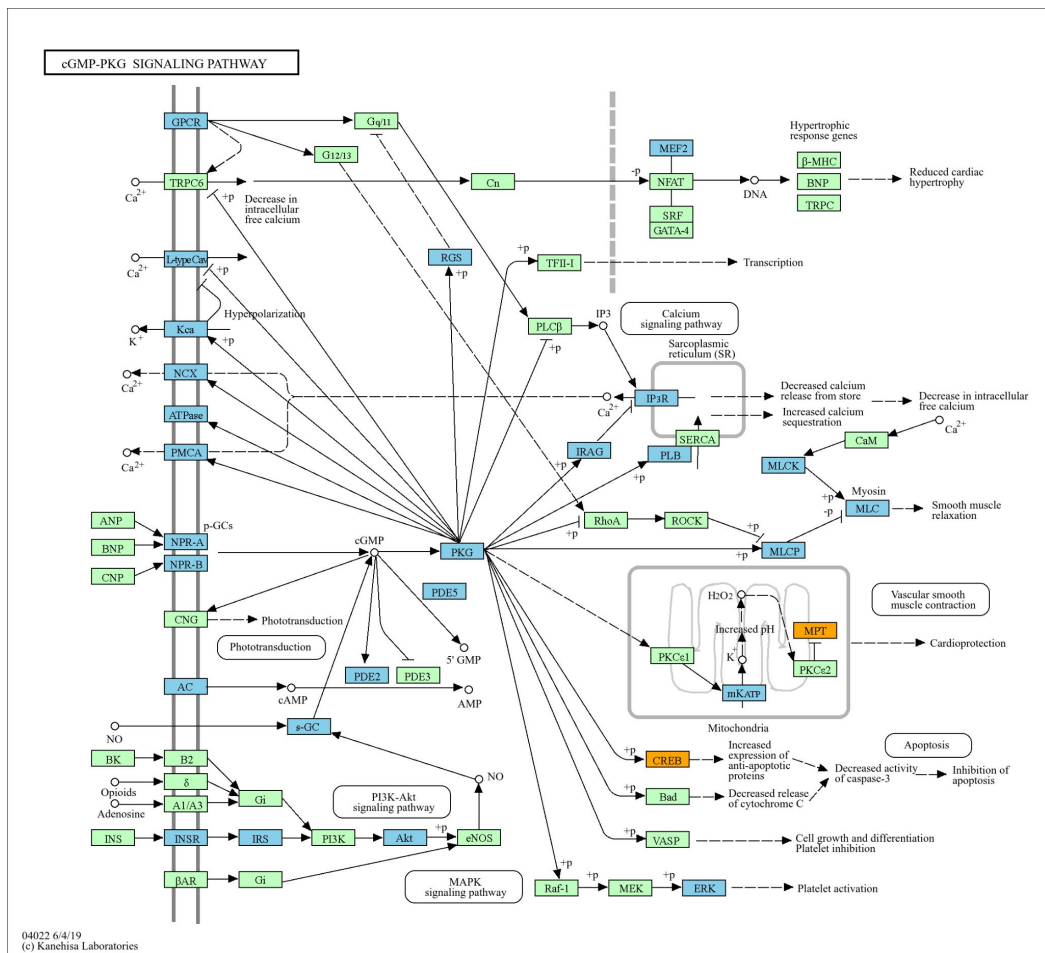
Cellular senescence is a state under which cells stop to replicate but remain metabolically active, as a response to stress factors or DNA damage. The process plays a key role in tumor suppression and it is typical to detect differential expression of related genes in cancer patients.

Moreover this article Wei C, Lin S, Huang Y, Wei Y, Mao J, Fan J. Integrated machine learning identifies a cellular senescence-related prognostic model to improve outcomes in uterine corpus endometrial carcinoma. Front Immunol. 2024 Jun 27;15:1418508. doi: 10.3389/fimmu.2024.1418508. PMID: 38994352; PMCID: PMC11236550. suggests that “targeting cellular senescence genes could be a promising personalized UCEC treatment”

KEGG/MAPK signaling



KEGG/cGMP-PKG signaling



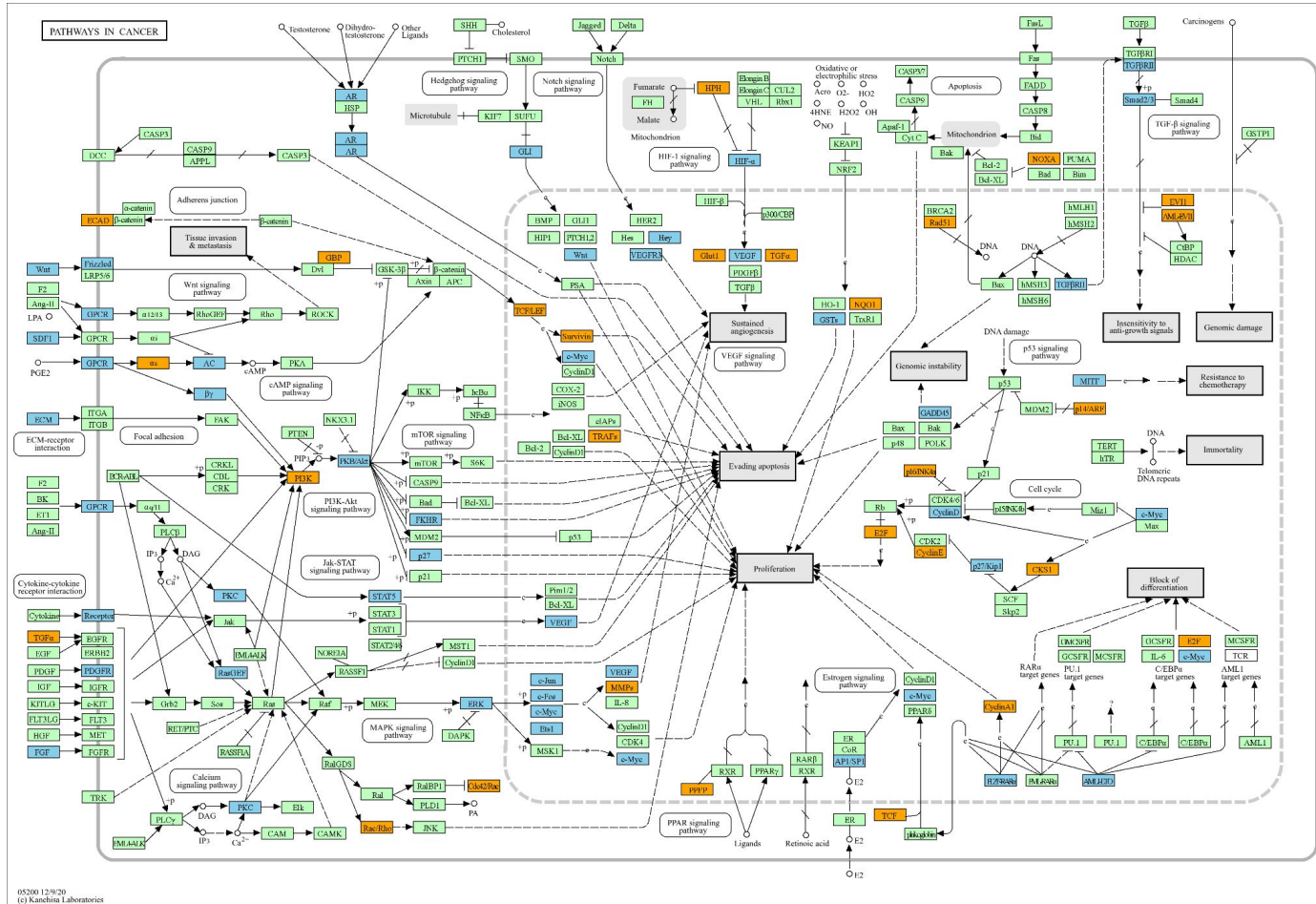
KEGG/ signaling pathways

Signaling (or signal transduction) is the process by which cells receive, process, and respond to information from their environment or from other cells.

It's how cells communicate and make decisions regarding growth, division, differentiation, movement, activation of genes and apoptosis.

We can clearly say that many of the involved genes are downregulated, concluding how their expression actually influences the processes and the progress of the disease is a far more complex matter.

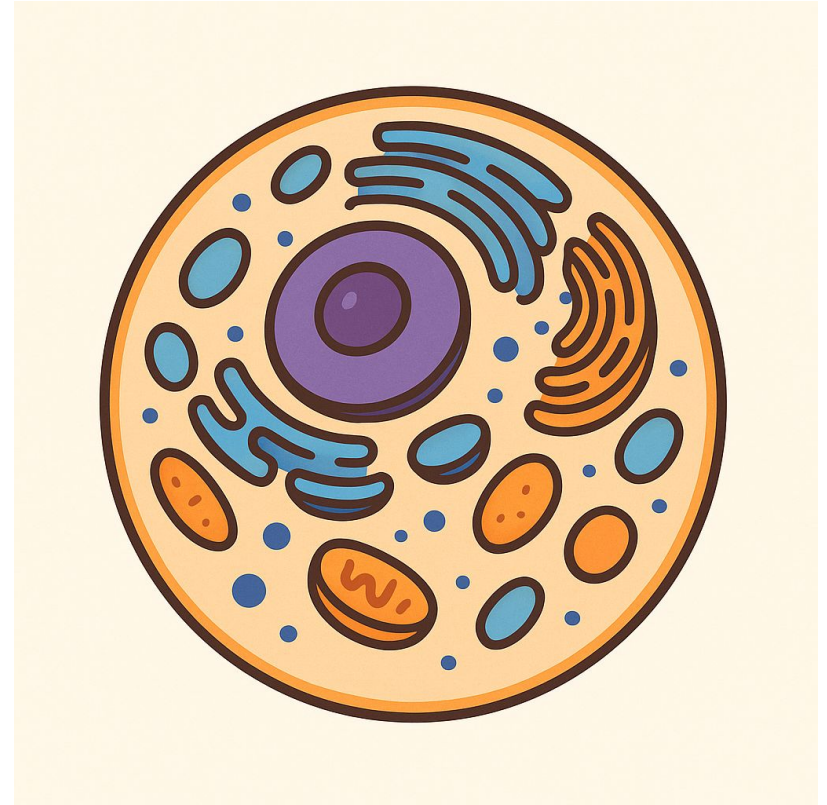
KEGG/Pathways in cancer



KEGG/Pathways in cancer

This is a more general pathway that did not explicitly showed up from the enrichment analysis but surely is a more general view on the topic.

We can clearly see that many differentiated genes are involved in processes such as: **apoptosis**, **proliferation** and many types of **signaling**.



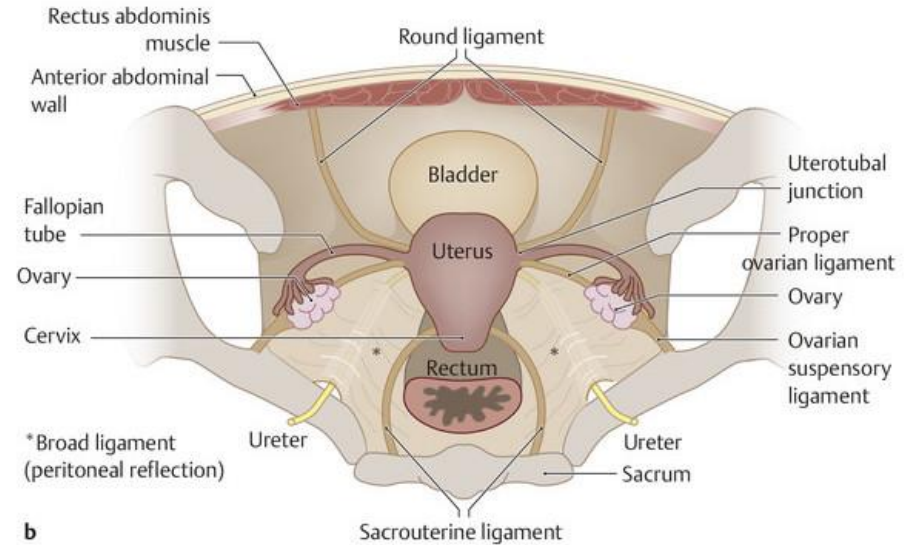
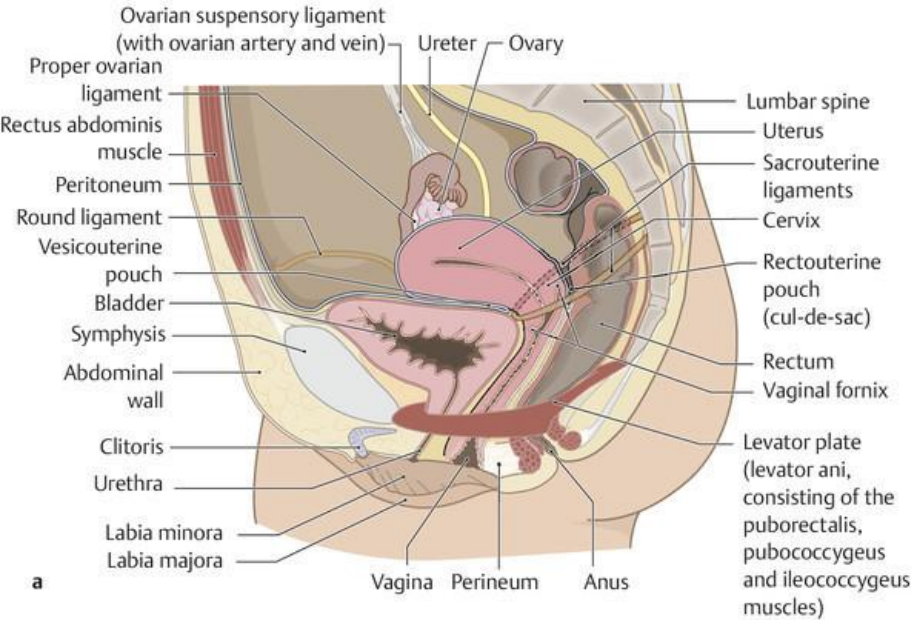
Further analysis

During the literature based search and after seeing the results of the enrichment (especially DisGeNET) I noticed that medical conditions concerning “surrounding” organs were mentioned very often.

In particular the ones that were mentioned the most are included in the so-called **pelvic area: ovaries, color, rectum.**

Since the subject of the analysis is a type of cancer that is known to be “aggressive” I performed further analysis about **metastasis.**

Female anatomy/pelvic area



Metastasis

With the term “**metastasis**” refers to the process by which cancer cells detach from the original site and invade surrounding tissues forming new tumors.

The “new” tumours can also arise from distant locations in the body since cancer cells can travel through the cardiovascular system and lymphatic system. This is possible since cancer cells can avoid detection by the **immune system**.

The term “cancer” itself refers to tumors that cause metastasis (malignant tumors, in contrast to benign tumors that do not invade nearby tissues but can still cause serious damage and be life threatening)

Metastasis/analysis

The first thing I did to confirm that the given samples had a component of metastasis was to analyze the **clinical_ucec.txt** file provided along with the gene expression matrix.

The second field of the file reports, for every patient, a **FIGO stage** value.

The FIGO (International Federation of Gynecology and Obstetrics) staging system is used for cancers of the female reproductive system. The roman numeral of the value (e.g Stage I, Stage II etc.) refers to the stage itself while the other literals indicate sub-stages within the main stage (e.g., Stage IA1, Stage IB2). These stages are used to classify the extent of cancer spread and predict treatment outcomes.

FIGO staging system

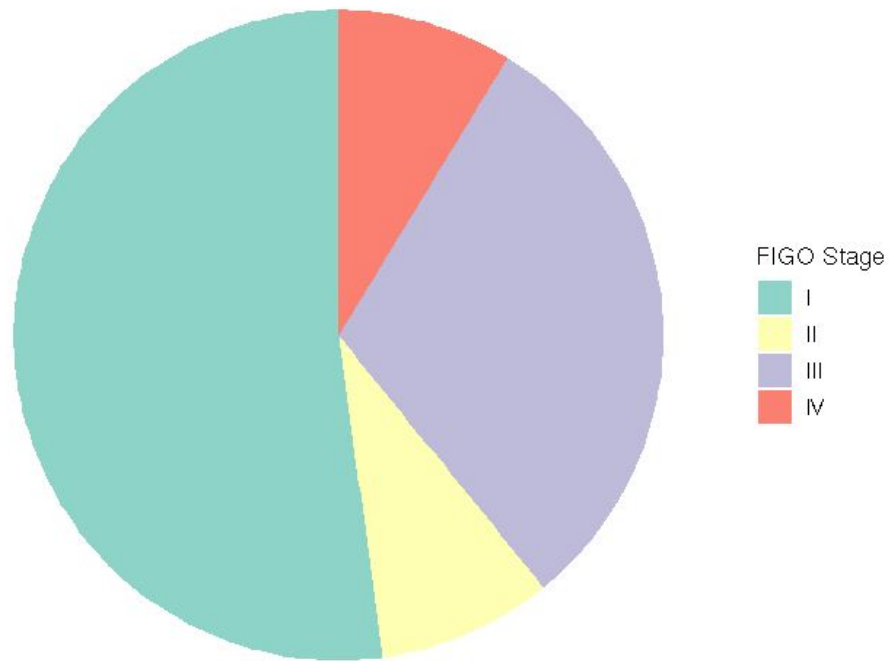
STAGE I	Confined to the uterine corpus and ovary
STAGE II	Invasion of cervical stroma without extrauterine extension OR with substantial lymphovascular system invasion OR aggressive histological types with myometrial invasion
STAGE III	Local and/or regional spread of the tumor of any histological subtype
STAGE IV	Spread to the bladder mucosa and/or intestinal mucosa and/or distant metastasis

FIGO stage/pie chart

Using an R script I was able to produce this pie chart that shows the distribution of FIGO stages in the dataset patients. (after filtering)

Notice that about 50% of the patients are in stage II or higher.

FIGO Stage Distribution



FIGO stage conclusions

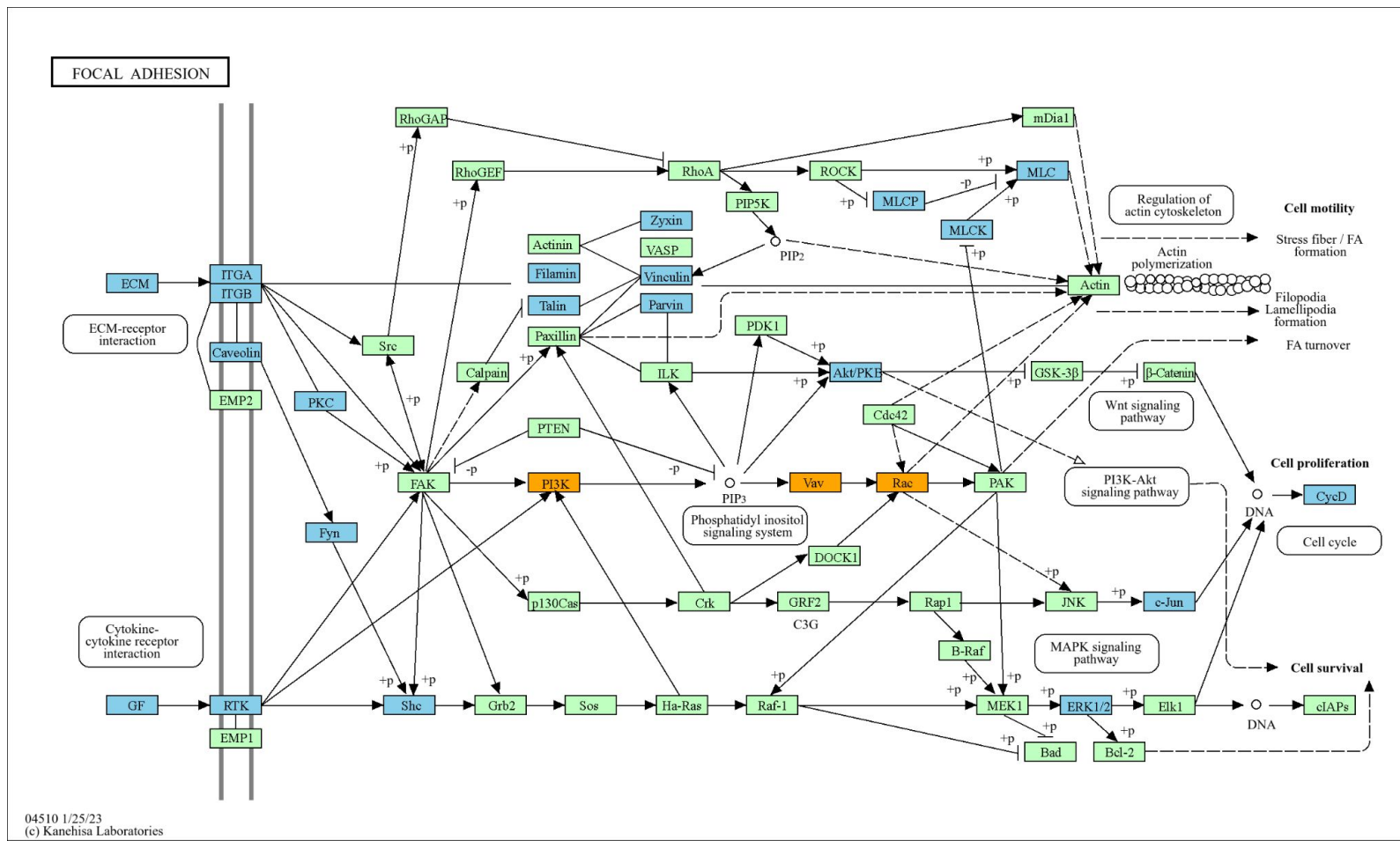
This data tells us that a consistent proportion of the patients has undergone a metastatic phase by spreading into nearby organs and structures.

Moreover, according to the FIGO classification, even a stage I endometrial cancer could be potentially associated with a spread to the ovaries.

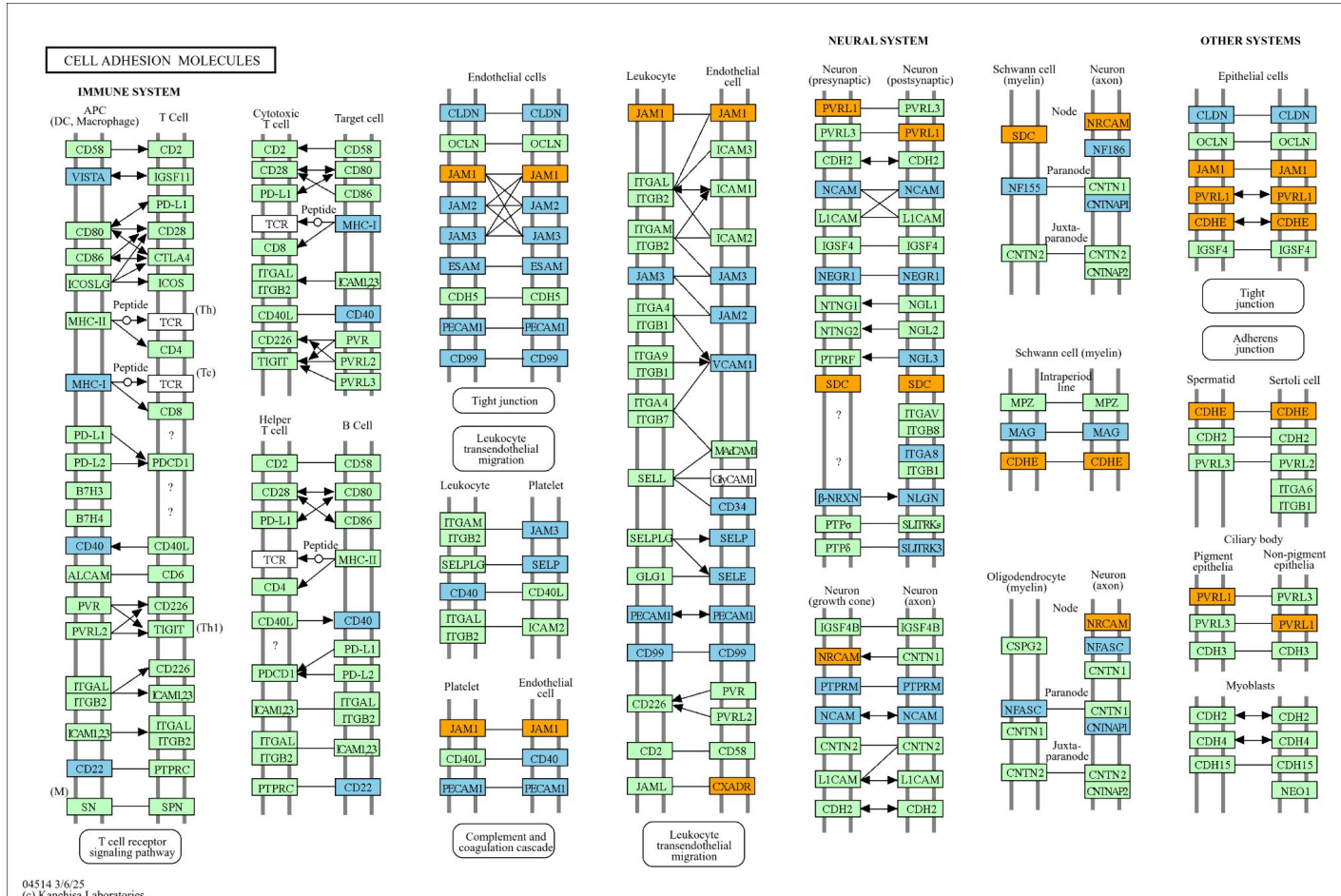
These results are in agreement with those found by the enrichment analysis, especially the ones from DisGeNET.

Some peculiar metastasis-related pathways were analyzed on KEGG.

Focal Adhesion



Cell Adhesion Molecules



Metastasis pathways

Focal adhesion refers to the processes and structures that are involved in cell anchoring to the extracellular matrix (ECM) while the **cell adhesion molecules** pathway puts a spotlight on the actual molecules involved in cell binding.

In general we can observe that a large number of genes involved are the ones that resulted to be downregulated in the analysis. These pathways point out the role of **binding** in metastasis both in the “detachment” phase from the original tumor and in the “reattachment” phase onto healthy tissue.

Thank you for your attention